

Computer Engineering Department

King Mongkut's University of Technology Thonburi

CPE 241 Database

Problem Session #09 User Interface Design for Database Systems

Date: Check PS09 assessment activities on LEB2

Introduction

In this lab, after refining your database design with business requirements and normalization, you will focus on how administrative users interact with the system. Instead of implementing a full-scale application, we will concentrate on designing the admin interface to facilitate system maintenance and management.

You will start by defining admin user scenarios, considering how they interact with the system to perform essential tasks. Based on these scenarios, you will design the user journey, mapping out the key interactions required for smooth operation. Following this, you will create UI designs that enable admins to efficiently manage the database.

Objectives

1. To analyze administrative workflows in a database system.
2. To design user interface structures that support database management tasks.

Instructions

STEP 1: Identify Admin User Scenarios

Define how administrative users interact with the system to maintain and manage database records. Then specify key administrative tasks. Examples of administrative tasks may include:

- Adding new records to the database
- Updating or correcting existing information
- Deleting obsolete records
- Generating reports for system monitoring or analysis

Clearly describe the scenario and the purpose of each task.

STEP 2: Design the User Journey (UX Design)

Map the workflow of an admin performing each task. Also, describe the sequence of steps required to complete the task, such as:

- Accessing the login page
- Navigating to the admin dashboard
- Selecting a management function
- Executing data operations

The workflow should demonstrate logical navigation and efficient interaction with the system.

STEP 3: Create UI Designs for Admin Functions

Develop UI mockups representing the key screens of the administrative system. Typical interface components may include:

- Login Page: secure access point for administrative users.
- Data Management Pages: interfaces for adding, editing, or deleting records.
- Reports and Analytics Section: interfaces for generating and viewing reports based on system data.

UI designs may be created using diagrams, wireframes, or design tools.

STEP 4: Interface Validation

Review the designed interface and ensure it supports the defined admin scenarios.

Verify that:

- All required administrative tasks can be performed.
- Navigation between screens is clear and logical.

Deliverables

- Documentation describing admin user scenarios.
 - User journey diagrams or workflow descriptions.
 - UI mockups for the main administrative interfaces.
 - Screenshots or images of all designed interface screens.
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Final Submission

- Compile all written explanations and pictures into a single a PDF file and ensure it is named appropriately (e.g., **PS09_YourGroupName .pdf**) then submit the report on LEB2 before the deadline.
- Upload the PDF report to the Assessment Activities section for PS09 on LEB2.